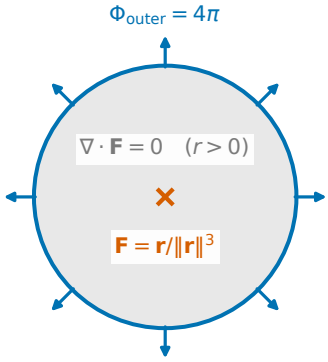
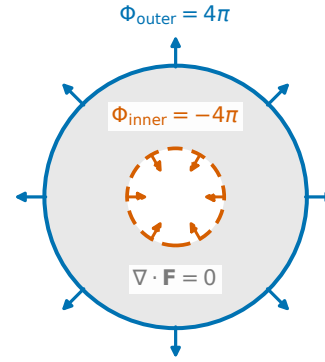
A



$\mathbf{F}(0)$ undefined

B



$\Phi_{\text{total}} = 4\pi - 4\pi = 0$